ANGLO-CHINESE JUNIOR COLLEGE

JC2 Preliminary Examination

Higher 1

GEOGRAPHY

8834/01

Paper 1

26 August 2024

3 hours

Additional Materials: Insert
Cover Sheet
Answer Paper

READ THESE INSTRUCTIONS FIRST

You should follow the instructions on the cover sheet. If you need additional answer paper, ask the invigilator.

Write your index number and name on the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer Question 1 in **Section A**.
Answer both questions in **Section B**.
Answer **two** questions in **Section C.**

Start your answer to each question on a fresh sheet of answer paper.

At the end of the examination, your answers should be separated and securely fastened into **2 bundles**. Attach the cover sheet on top of Bundle 1.

- Bundle 1: Cover sheet and answers to Questions 1 and 2.
- Bundle 2: Answers to Question 3 and Questions in Section C.

The number of marks is given in brackets [ ] at the end of each question or part question.

This document consists of **4** printed pages.

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[Turn over

Section A

Cluster 2 Fieldwork

1 A group of students conducted a fieldwork investigation on the accessibility of transportation and quality of outdoor spaces for the elderly in Bukit Merah. The focus of their investigation was to assess how well the community's infrastructure supports the elderly in staying active and independent. The hypothesis for their study was:

'Public transportation and outdoor spaces in Bukit Merah do not adequately meet the needs of elderly residents.'

The students conducted the investigation over two weekdays in August, between 9am and 12pm. The two main locations selected were the Bukit Merah Bus Interchange and the Tiong Bahru Plaza shopping mall. These locations were chosen based on their accessibility and frequency of use by elderly residents. The students decided to collect data by conducting a survey and taking photographs of the area.

A total of 30 elderly respondents, aged 60 and above, participated in the survey. The aim was to gather data on their experiences with transportation and outdoor spaces.

Resource 1 shows photographs of Bukit Merah Bus Interchange and Tiong Bahru Plaza shopping mall. Resource 2 shows the survey that the students used. Resource 3 shows selected results of the survey.

(a) Explain why this hypothesis is appropriate for the investigation. [4]

Point marked
Award 1 mark for each explanation on suitability of students' hypothesis, to a maximum of 4 marks.
Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Clearly defined: The hypothesis is clearly defined, focusing on two variables—public transportation and outdoor spaces—and a specific population group, the elderly residents in Bukit Merah.

Additional mark: The focus on elderly residents highlights a key demographic group in the community that may have unique needs, making the investigation both focused and relevant.

Scale: The selected locations, Bukit Merah Bus Interchange and Tiong Bahru Plaza, are appropriate as they are commonly used by the elderly and represent important spaces for assessing accessibility and outdoor environment quality.

Additional mark: These locations are easily accessible, allowing students to collect data efficiently over the two days. The proximity between locations ensures consistency in data collection without major spatial variability.

Scale: Conducting the investigation over two weekdays between 9 am and 12 pm is appropriate, as these times are likely to capture elderly residents using public transportation and outdoor spaces for morning activities.

Additional mark: The chosen timeframe coincides with non-peak hours, which may provide a better understanding of elderly-specific usage patterns in both public transport and outdoor spaces.

Methods: The use of surveys is suitable for gathering subjective data about the experiences of the elderly with transportation and outdoor spaces, providing direct insight into whether their needs are being met.

Additional mark: The inclusion of photographs enhances the investigation by allowing the students to document physical infrastructure, supporting the survey findings with visual evidence.

(b) Describe how the students can improve the sampling method used in their investigation. [3]

Award 1 mark for each description

Increase sample size: The sample size of 30 elderly respondents may not be sufficient to provide a representative view of the elderly population in Bukit Merah. Increasing the number of participants would improve the reliability of the results.

Use stratified sampling: The students could employ a stratified sampling method to ensure different subgroups of the elderly population (e.g., age groups, mobility levels, or those living in different areas of Bukit Merah) are included.

Survey at different times of day: Conducting the survey only between 9 am and 12 pm might not capture the experiences of elderly residents who use public transportation or outdoor spaces at other times. The students could survey at different times of the day, including afternoons and evenings.

(c) With reference to Resources 1 and 2, explain two possible issues related to ethics when students collect primary data. [4]

Point marked
Award 1 mark for each explanation on each issue related to ethics, to a maximum of 4 marks.
Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Informed consent: The elderly respondents may not fully understand the purpose of the survey(as shown in Resource 2), especially if it wasn't clearly explained to them. This could lead to participants providing data without being fully aware of how it will be used.

Additional mark: To ensure informed consent, the students should clearly explain the objectives of the investigation and how the information will be used before collecting any data. This can be done through an information sheet or verbal explanation.

Anonymity and privacy: The students may have taken photographs of the elderly in public spaces (as shown in Resource 1) without considering their privacy or asking for permission. This raises concerns about protecting the identity of the participants.

Additional mark: The students should blur or avoid capturing identifiable features of the elderly individuals in photographs or seek explicit permission to photograph them. Additionally, personal data collected in the survey should be kept anonymous to maintain privacy.

(d) With reference to Resource 1, suggest how the students could minimise potential risks when they undertake fieldwork at Bukit Merah Bus Interchange and Tiong Bahru Plaza shopping mall. [6]

Point marked
Award 1 mark for each risk described with reference to Resource 1.
Award a maximum of 1 additional mark for each suggestion to minimize the risk where applicable.

Minimising risks in crowded areas: Bukit Merah Bus Interchange and Tiong Bahru Plaza are likely to be crowded, which could pose risks such as accidents or loss of personal belongings.

Additional mark: The students could wear visible identifiers, such as vests, to make themselves easily recognizable, and conduct their surveys in safe, designated areas away from heavy foot traffic to avoid accidents. Students should be cautious while moving through busy areas and avoid obstructing walkways when conducting surveys or taking photographs.

Safety in public spaces: As both locations are public, students may face risks from interacting with unfamiliar individuals. To mitigate this, they should work in pairs or groups, ensuring that no one is collecting data alone.

Additional mark: Students should also inform their teacher or supervisor of their whereabouts at all times and carry a mobile phone in case they need assistance or face emergencies.

Traffic hazards: Since Bukit Merah Bus Interchange is a busy transportation hub, students may be exposed to the risk of accidents from moving buses or vehicles while conducting their fieldwork.

Additional marl: To minimize this risk, students should stay clear of bus lanes and only conduct surveys or take photographs in designated pedestrian areas or safe locations.

Disruption to public spaces: Conducting surveys in busy areas like Tiong Bahru Plaza may disrupt the flow of people or lead to conflicts with shop owners or security staff if the students are seen as obstructing business operations.

Additional mark: To avoid this, the students should seek permission from mall management before conducting surveys and ensure they do not block entrances or create bottlenecks in high-traffic areas.

(e) With reference to Resource 3, explain the limitations of the data representation shown. [3]

Point marked
Award 1 mark for each explanation on limitations of data representation, to a maximum of 4 marks.
Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Lack of visual appeal: Data presented in a table can be overwhelming and difficult to interpret at a glance, especially for those unfamiliar with data analysis. It may not effectively highlight key patterns or trends in the results, making it less accessible for readers.

Difficulty in comparing data: Tables may require the reader to manually scan and compare values across rows and columns, which can be time-consuming and prone to errors. This makes it harder to draw clear conclusions from the data.

(f) Evaluate the validity of the data shown in Resource 3 in answering the hypothesis. [10]

Indicative Content

Candidates can evaluate the usefulness of the data in Resource 3 by examining how well it addresses the hypothesis that public transportation and outdoor spaces in Bukit Merah do not adequately meet the needs of elderly residents. One approach would be to focus on the sample size and representativeness of the data. With only 30 respondents surveyed at two specific locations (Bukit Merah Bus Interchange and Tiong Bahru Plaza), the data may not be large or diverse enough to generalize the experiences of the elderly population in Bukit Merah as a whole. The students' choice of conducting surveys during limited hours on weekdays further narrows the scope, possibly missing elderly individuals who use public spaces or transportation at different times. Another approach would be to assess the data collection methods, particularly the reliance on quantitative data and the lack of qualitative insights. While the numerical data in the table provides a basic overview, it does not capture the complexity of elderly residents' experiences, such as specific challenges related to accessibility or mobility.

Candidates should also consider the limitations of the investigation due to its reliance on the survey method, which may not fully capture the physical infrastructure issues of public transportation and outdoor spaces. For instance, the survey does not account for external factors such as weather conditions or the condition of facilities like ramps, benches, and shaded areas. Furthermore, the survey's wording may have used subjective terms that could lead to misunderstanding among elderly respondents, especially if there were any language barriers. In terms of geographical representation, the investigation focused on two main locations, leaving out other parts of Bukit Merah that may offer different infrastructural challenges or opportunities. The lack of temporal variation in the data collection (only two days during a short window of time) also undermines its ability to fully assess the adequacy of transportation and outdoor spaces, as the elderly may use these facilities differently on weekends or during peak hours.

Section B

Cluster 1: Sustainable Future and Climate Change

Question	Answer	Marks
2(a)	**Describe the pattern of temperature anomalies shown in Resource 4**. Any one of the following, up to a max. of 4 marks. • Highest temperature anomaly increase observed in the high latitudes [1 mark] In locations like the Arctic temperature is observed to have increased by more than 2.5^{o}C, while northern Canada and northern Europe have shown temperature increase between 1.5 to 2.5^{o}C [1 additional mark] • In the tropics, temperature anomalous increase range from 1.5 to 2°C [1 mark] • Ocean in the northern hemisphere show higher temperature increase of 1 to1.5^{o}C, compared with the southern hemisphere oceans ranging from 0.3 to 1°C [1 mark] The waters around Southeast Asia and in the South India Ocean showed increases but around 0.5 to1^{o}C [1 additional mark] • There is a distinct decrease in temperature in the north Atlantic Ocean just south of Greenland and Iceland. [1 mark]	[4]

Question	Answer	Marks
2(b)	**Compare the patterns of CO_2 emissions from the top ten countries shown in Resource 5.** Any one of the following, one mark each, up to a max. of 4 marks. • Of the top ten countries in terms of CO_2 emission, six are high-income or developed countries, while four are low-income or developing countries including China, Brazil, Indonesia and India. • The bulk of the CO_2 emissions from the developed countries arise from the combustion of fossil fuels. For example in the case of USA which rank first, 420 billion tonnes come from the combustion of fossil fuel while 85 billion tonnes from land use change and loss of forests. • In contrast, developing countries like Brazil and Indonesia, which rank 4th and 5th respectively in term of cumulative CO2 emissions from 1850 to 2021, larger proportion of close to 90% arise from land use change and loss of forest cover. • China, the 2nd largest emitter of cumulative CO2 is a developing country where fossil fuel emission contributed a larger portion compared to land use change and loss of forest cover. • One developed country, ie, Canada showed almost equal contribution from fossil fuel and land use change. • Germany is exception in the top ten countries in terms of cumulative CO2 emission where there is a negative in emission, ie, the is a decrease in emission in the area of land use change, and instead there is an increase in the sink for CO2, albeit at a low value of 0.5 billion tonnes in the entire period from 1851 to 2021. • UK and Japan are two developed countries that have the bulk of their CO2 emission from fossil fuel combustion and use compared to that from change of land use and forest cover.	[4]

Question	Answer	Marks
2(c)	**Explain why the areas showing the greatest increase in temperature in Resource 4 do not necessarily correspond to the countries with greatest CO_2 emissions shown in Resource 5.** Responses will focus on the high temperatures in the Arctic area and the northern parts of Europe and the ice-albedo feedback • Greenhouses gases (GHGs) generated by the countries like USA and China will flow with the atmospheric circulation, and is not confined to the locations that emit the GHGs. • High GHGs in the atmosphere leads to the enhanced greenhouse effect leading to higher average temperature globally. • In locations with snow cover and icesheets particularly, the high reflectivity or albedo of the location is reduced when the high temperature melts the icesheets. With the lower cover of reflective icesheets and more of the darker surface exposed at the surface to absorb the incoming solar radiation, temperature right further there. With further temperature increase, more icesheets melt, and the positive feedback loop reinforces the chain reaction of rising temperature, melting icesheets, decreasing albedo and further temperature. • Other locations without the icesheets and albedo effect do undergo temperature increase, but without the mechanism of the ice-albedo feedback do not show such a large increase in temperature.	[4]

Question	Answer	Marks
2(d)	**Explain two arguments that support the view that contemporary climate change is caused by human activities**. Any of two of the following are possible responses • **Increase in Greenhouse Gas Emissions**: One of the most compelling arguments is the significant rise in greenhouse gases, particularly carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O), due to human activities. Since the Industrial Revolution, the burning of fossil fuels (such as coal, oil, and natural gas) for energy, deforestation, and industrial processes have dramatically increased the concentration of these gases in the atmosphere. Scientific measurements show that CO_2 levels have risen from about 280 parts per million (ppm) before the industrial era to over 410 ppm and were higher than at any time in at least 2 million years, and concentrations of methane (1866 parts per billion) and nitrous oxide (332 parts per billion) were higher than at any time in at least 800,000 years. These increases correlates strongly with the rise in global temperatures, as greenhouse gases trap heat in the Earth's atmosphere, leading to the warming effect known as the enhanced greenhouse effect. • **Isotopic Evidence** Furthermore, the chemical composition of manmade CO_2 is more abundant than the naturally formed CO_2 from the natural carbon cycle. The carbon in CO_2 from fossil fuels has a distinct isotopic signature, different from that of carbon from natural sources. The increasing presence of this specific isotopic form of carbon in the atmosphere is direct evidence that the additional CO_2 is from burning fossil fuels, not from natural processes. • **Climate Models**: Scientists use sophisticated climate models to simulate the Earth's climate. These models incorporate both natural factors (like solar radiation and volcanic activity) and human activities (such as greenhouse gas emissions and deforestation). When only natural factors are included, the models fail to replicate the warming observed over the past century. Natural factors alone cannot account for the observed warming, particularly the sharp rise in global temperatures over the last century. However, when human activities are added to the models, the simulated results closely match the observed temperature increases. This indicates that human activities are the primary driver of the recent warming.	[4]

Question	Answer	Marks
2(e)	Describe two natural causes of climate change leading to glacial and inter-glacial periods. **1. Milankovitch Cycles** Milankovitch cycles are variations in Earth's orbit and axial tilt that occur over tens of thousands to hundreds of thousands of years. These cycles influence the distribution and intensity of sunlight reaching Earth, which in turn affects the planet's climate. There are three main types of Milankovitch orbital cycles: • **Eccentricity:** This cycle refers to changes in the shape of Earth's orbit around the Sun, from more circular to more elliptical, occurring over about 100,000 years. When the orbit is more elliptical, there is a greater difference in solar energy received by Earth between the closest point (perihelion) and the farthest point (aphelion). This can lead to significant climate changes, including the initiation of glacial periods when less solar energy reaches the planet. • **Axial Tilt (Obliquity):** The angle of Earth's axis relative to its orbital plane changes over a cycle of about 41,000 years. When the tilt is greater, the seasons become more extreme, with hotter summers and colder winters. Changes in axial tilt can influence the advance and retreat of ice sheets, contributing to glacial and interglacial periods. Larger tilt angles favour periods of deglaciation (the melting and retreat of glaciers and ice sheets). Smaller tilt results in summer that is not able to melt the previous winter snow cover, increasing albedo, and further increasing icesheet growth. • **Precession:** This is the wobble in Earth's axis as it rotates, occurring over approximately 26,000 years. Precession affects the timing of the seasons in relation to Earth's position in its orbit. This can amplify or reduce seasonal contrasts, affecting ice sheet growth or melting, and thereby contributing to glacial and interglacial cycles. When northern hemisphere summer occurs when Earth is at perihelion (closest location in relation to the sun) the higher insolation can set in the interglacial period. **2. Volcanic Activity** • **Volcanic Eruptions**: Large volcanic eruptions can inject massive amounts of aerosols, particularly sulfur dioxide (SO_2), into the stratosphere. These aerosols form sulfate particles that reflect sunlight back into space, leading to temporary cooling of the Earth's surface. • **Long-Term Impacts**: While individual eruptions typically cause short-term cooling (lasting from months to a few years), repeated or particularly large volcanic eruptions over extended periods can contribute to longer-term climate shifts. For example, sustained volcanic activity could contribute to cooler conditions that might help trigger the expansion of ice sheets during a glacial period. • **Greenhouse Gas Release**: On longer timescales, volcanic activity also releases greenhouse gases like CO_2. However, during periods of high volcanic activity, the cooling effects of aerosols can dominate over the warming effect of CO_2, contributing to glacial conditions. Conversely, a reduction in volcanic activity could contribute to interglacial warming by allowing CO_2 levels to rise.	[6]

	These natural processes have played significant roles in the Earth's climate system over geological timescales, leading to the cyclical advance and retreat of ice sheets that characterize glacial and interglacial periods.	

3 Resource 6 shows photographs of the London docklands in 1968 and 2018. Resource 7 shows the population changes in the London docklands towns from 1801 to 2019. Resource 8 shows an article about Canary Wharf.

(a) Using Resource 6, describe the changes in the London docklands from 1968 to 2018 . [4]

Point marked

In 2018, the London Docklands is dominated by modern, glass skyscrapers compared to the previous industrial area.
In 2018, there are green spaces that break up the dense urban environment, providing open spaces that enhance the aesthetic and livability of the district.
In 2018, there is land reclamation done with land expanded to build the modern buildings seen.
In 2018, there is greater transport infrastructure put in with highways and roads leading into the docklands

(b) Explain the economic reasons for urban reimaging efforts. [4]

Point marked
1 mark for each reason and 1 mark for further development.

Attracting Investment: Urban reimaging efforts are often driven by the need to attract both domestic and foreign investments. By redeveloping and modernizing urban areas, cities can present themselves as attractive destinations for businesses, which in turn generates economic growth through job creation and increased tax revenues.

Boosting Tourism: Another economic reason is to boost tourism by transforming the city's image and infrastructure to appeal to visitors. By enhancing cultural landmarks, creating visually appealing public spaces, and modernizing transportation, cities can attract more tourists, leading to increased spending in local businesses and the hospitality sector.

(c) Describe the changes in population in the London Dockland towns from 1801 to 2019 as shown in Resource 7. [4]

Point marked
1 mark for each change described.

All three dockland towns experience similar trends in population from 1801 to 2019.
All three towns had an increase from 1801 to 1901, however, Newham grew sharply only from 1861 onwards compared to Southwark and Tower Hamlets which had a steady increase from 120000 to 600000.
All three towns also experienced a decrease in population from 1901 onwards to 1981 with Southwark going from 600000 to 200000.
All three then experience an increase in population from 1981 to 2019.

(d) With reference to Resource 8, explain the impacts of urban reimaging on sustainable urban development. [6]

Point marked
1 mark for each impact identified and 1 mark for further development

Economic Sustainability: The transformation of Canary Wharf into a global financial hub has generated economic growth and job opportunities. The presence of major corporations like Barclays, HSBC, and JP Morgan has attracted investment, creating a stable and prosperous economy in the area, which supports long-term financial sustainability.

Social Sustainability: The development of housing projects like Wood Wharf, which brings 4,000 new homes, helps address housing shortages, providing more residential spaces for a growing population. By incorporating a "green spine" in partnership with the Eden Project, the development also ensures that residents have access to green spaces, contributing to the overall quality of life and well-being of the community.

Environmental Sustainability: The introduction of green spaces, such as the "green spine" and other open areas, contributes to a more sustainable environment by improving air quality and providing urban biodiversity. Additionally, infrastructure projects like the Docklands Light Railway (DLR), Jubilee Line extension, and the new Elizabeth Line promote sustainable transportation, reducing the reliance on cars and lowering carbon emissions.

(e) Explain the tensions between sustainable urban development and liveability. [4]

Point marked
1 mark for each tension explained and 1 mark for further development

Environmental sustainability efforts, such as eco-friendly infrastructure and green spaces, can raise property values, making housing less affordable and reducing liveability for lower-income residents. As housing costs rise, many are forced to move away from sustainable urban areas, creating a tension between maintaining affordability and achieving environmental goals.

Promoting higher urban density to reduce sprawl supports sustainability, but it can lead to overcrowding and strain on public services. This can negatively impact liveability by reducing access to amenities, increasing congestion, and creating pressure on transportation and public spaces, diminishing the quality of life for residents.

Section C

Answer **two** questions

4 'Rapid urban growth is the greatest challenge for countries progressing towards sustainable urban development.'

Evaluate this statement. [13]

Indicative content

Rapid urban growth leads to challenges such as environmental degradation, overpopulation, and increased pressure on resources, all of which hinder sustainable urban development. In cities like Jakarta, informal settlements and slums emerge as urban populations grow faster than infrastructure can keep up. This creates problems such as water shortages, pollution, and inadequate housing, which undermine the goals of sustainable development. Rapid urbanization can also cause a strain on transportation systems and contribute to worsening air quality, impacting both liveability and environmental sustainability.

While rapid urban growth presents challenges, it is not the sole or even greatest obstacle to sustainable development. Political, economic, and institutional barriers such as poor governance, lack of funding, and corruption may pose larger hurdles. Some cities, such as Curitiba in Brazil, have successfully managed rapid urbanization by implementing effective policies on waste management, green spaces, and integrated public transportation systems. This suggests that with proper governance and urban planning, rapid urban growth can be managed in a way that supports sustainability.

The extent to which rapid urban growth is the greatest challenge depends on the local context and governance structures. In developing countries where infrastructure development lags behind population growth, it may indeed be the primary challenge. However, in more developed contexts, challenges like economic inequality, political instability, or climate-related issues could pose more significant barriers to achieving sustainable urban development.

5 'Climate change affects all regions equally.'

Evaluate this statement. [13]

Indicative content

Climate change is a global phenomenon that has caused rising temperatures, changing weather patterns, and more frequent natural hazards across the world. Regions in both developed and developing countries experience impacts such as extreme heat, floods, and changing agricultural conditions. For example, the global rise in sea levels is a challenge faced by both low-lying coastal regions in the UK and Pacific Island nations, suggesting that climate change impacts are widespread and affect all areas to some degree.

Climate change affects different regions in varying degrees due to geographical location, socio-economic status, and adaptive capacity. Vulnerable regions like Sub-Saharan Africa or small island states experience disproportionately severe impacts due to weaker resilience and greater exposure to extreme weather events. Developed nations, such as those in Europe or North America, tend to have greater financial and technological

resources to adapt and mitigate the effects of climate change, while poorer regions suffer more from food insecurity, health risks, and displacement. For example, Bangladesh faces much greater impacts from flooding than a country like Japan, which has better infrastructure for dealing with such events.

While climate change is a global issue, the impacts are unevenly distributed. Regions with less adaptive capacity and greater exposure to climate hazards are more severely affected, whereas wealthier regions with more resources are better equipped to mitigate these impacts. This makes the statement inaccurate, as the geographical context plays a critical role in determining the severity of climate change effects.

6 'Strategies to respond to climate change have largely been effective.'

Evaluate this statement. [13]

Indicative content

Several strategies to mitigate climate change, such as carbon trading, renewable energy adoption, and reforestation, have shown success. Countries like Sweden and Germany have significantly reduced their carbon emissions through the use of renewable energy sources like wind and solar. Global initiatives such as the Paris Agreement provide a framework for international cooperation on climate change mitigation, with many nations making commitments to reduce their carbon footprints and implement green technologies. These strategies show that progress is being made towards limiting climate change.

Despite these efforts, the overall effectiveness of global strategies to combat climate change has been limited. Global carbon emissions continue to rise, and many countries have failed to meet the targets set under international agreements like the Paris Agreement. Developing nations, such as India and Brazil, struggle to balance economic growth with environmental sustainability, and face challenges like deforestation and high fossil fuel dependence. Additionally, some mitigation strategies, like carbon offsetting, have been criticized for not addressing the root causes of emissions, and adaptation measures in vulnerable regions, such as sea walls in Bangladesh, are often underfunded and insufficient to fully protect communities from climate hazards.

While there have been localized successes in some developed countries and regions, the global response to climate change remains insufficient, particularly in the developing world. The uneven implementation of strategies, combined with rising global emissions, suggests that while progress has been made, these strategies have not yet been effective on a large enough scale to address the global challenge of climate change.